

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE-MEF-I (B)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR.
For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

Test Booklet Code

This Booklet contains **28** pages.

E4

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - (b) **Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
3. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720.**
4. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
5. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
6. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
7. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
8. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
9. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
10. No candidate, without special permission of the Invigilator, would leave his/her seat.
11. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
12. Use of Electronic/Manual Calculator is prohibited.
13. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
14. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
15. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
16. Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ **Invigilator's Signature :** _____

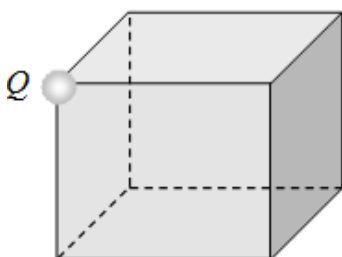
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SECTION-A (PHYSICS)

1. The circular scale of a micrometer has 200 divisions and pitch of 2mm. Find the measured value of thickness of a thin sheet.

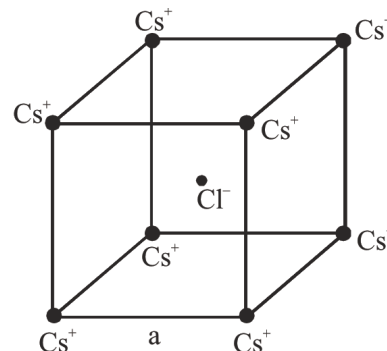


- (1) 3.41 mm (2) 6.41 mm
(3) 3.46 mm (4) 3.51 mm
2. Which of the following sets can't enter into the list of fundamental quantities in any system of units?
- (1) length, mass and velocity
(2) mass, time and velocity
(3) length, time and velocity
(4) length, time and mass
3. What is the number of significant figures in 0.310.
- (1) 2 (2) 3 (3) 4 (4) 6
4. A wire has mass $(0.6 \pm 0.006)\text{g}$, radius $(1 \pm 0.01)\text{mm}$ and length $(2 \pm 0.02)\text{cm}$. The percentage error in the measurement of its density will be
- (1) 4% (2) 6% (3) 3% (4) 7%
5. A charge Q is placed at any one vertex of a cube then flux passing from any one of the opposite faces of cube is -

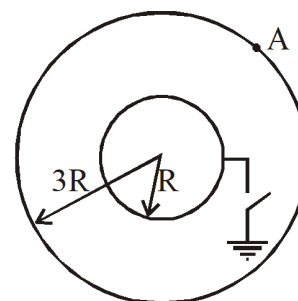


- (1) $\frac{Q}{6\epsilon_0}$ (2) $\frac{Q}{8\epsilon_0}$
(3) $\frac{Q}{24\epsilon_0}$ (4) $\frac{Q}{2\epsilon_0}$

6. In the basic CsCl crystal structure, Cs^+ and Cl^- ions are arranged in a bcc configuration as shown in the figure. The net electrostatic force exerted by the eight Cs^+ ions on the Cl^- ions is :-



- (1) $\frac{1}{4\pi\epsilon_0} \frac{4e^2}{3a^2}$ (2) $\frac{1}{4\pi\epsilon_0} \frac{16e^2}{3a^2}$
(3) $\frac{1}{4\pi\epsilon_0} \frac{32e^2}{3a^2}$ (4) zero
7. Concentric metallic shells of radii R and $3R$ are shown. Charge on the outer shell is q , find charge on inner shell, after closing the switch (before closing the switch, inner shell is neutral) :-



- (1) $+q$ (2) $-q$ (3) $+\frac{q}{3}$ (4) $-\frac{q}{3}$
8. On a conductor of nonuniform curvature, the charge :-
- (1) Has the greatest concentration on the part of greatest radius of curvature.
(2) Has the greatest concentration on the parts of least radius of curvature.
(3) Is distributed uniformly on the whole surface.
(4) Is distributed uniformly over its volume.

9. **Assertion** :- Under electrostatic conditions net electric field inside a solid conductor will be zero.

Reason :- under electrostatic conditions there will be no free electrons inside a conductor.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (3) Assertion is True but the Reason is False.
 (4) Both Assertion & Reason are False.

10. Choose incorrect statement/s :-

- (A) In friction (method of charging), body having low work function will acquire positive charge.
 (B) In conduction charge transfer takes place till potential of both bodies becomes equal.
 (C) Inducing body will get opposite nature of charge.
 (D) Induction is a particle phenomena.

- (1) Only A (2) Only B
 (3) Both A and B (4) Both C and D

11. The electric potential at a point (x, y, z) is given by

$$V = -x^2y - xz^3 + 4$$

The electric field E at the point is :-

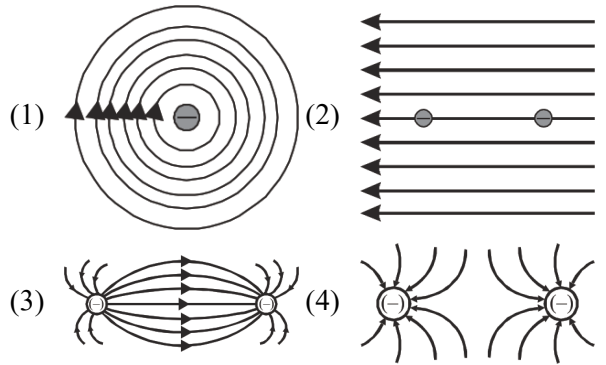
- (1) $\vec{E} = \hat{i}(2xy + z^3) + \hat{j}x^2 + \hat{k}3xz^2$
 (2) $\vec{E} = \hat{i}2xy + \hat{j}(x^2 + y^2) + \hat{k}(3xz - y^2)$
 (3) $\vec{E} = \hat{i}z^3 + \hat{j}xyz + \hat{k}z^2$
 (4) $\vec{E} = \hat{i}(2xy - z^3) + \hat{j}xy^2 + \hat{k}3z^2x$

12. The inward and outward electric flux for a closed surface in units of $\text{N}\cdot\text{m}^2/\text{C}$ are respectively 8×10^3 and 4×10^3 . Then the total charge inside the surface is :-

[where ϵ_0 = permittivity constant]

- (1) $4 \times 10^3 \text{ C}$ (2) $-4 \times 10^3 \text{ C}$
 (3) $\frac{(-4 \times 10^3)}{\epsilon_0} \text{ C}$ (4) $-4 \times 10^3 \epsilon_0 \text{ C}$

13. Which of the following represents the electric field lines due to a combinations of two negative charges :-



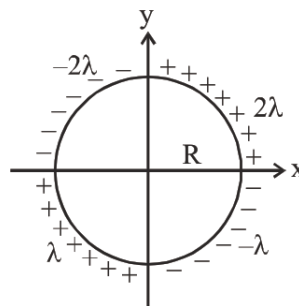
14. The electric field, at a distance of 20 cm from the surface and outside of a dielectric sphere of radius 10 cm is, 100 V/m. Then E at 3 cm distance from the centre of sphere is :-

- (1) 100 V/m (2) 125 V/m
 (3) 120 V/m (4) 270 V/m

15. Two charges each of magnitude Q are fixed at 4a distance apart. A third charge ((-Q) of mass 'm') is placed at the mid point of the two charge. now (-Q) charge is slightly displaced perpendicular to the line joining the charges then find its time period :

- (1) $T = 2\pi\sqrt{\frac{2m\pi\epsilon_0 a^3}{Q^2}}$ (2) $T = 2\pi\sqrt{\frac{4m\pi\epsilon_0 a^3}{Q^2}}$
 (3) $T = 2\pi\sqrt{\frac{8m\pi\epsilon_0 a^3}{Q^2}}$ (4) $T = 8\pi\sqrt{\frac{m\pi\epsilon_0 a^3}{Q^2}}$

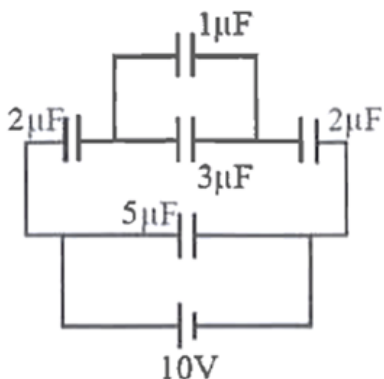
16.



As shown in figure. The magnitude of electric field at the centre is.

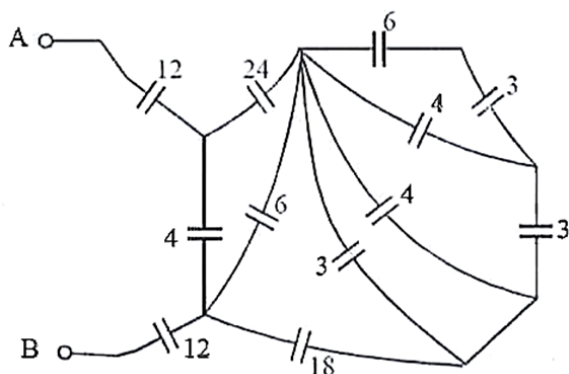
- (1) $\frac{\lambda}{2\pi\epsilon_0 \cdot R}$ (2) $\frac{\lambda}{\pi\epsilon_0 \cdot R}$
 (3) $\frac{\sqrt{2}\lambda}{2\pi\epsilon_0 \cdot R}$ (4) None of these

17. The ratio of potential difference between $1 \mu\text{F}$ and $5 \mu\text{F}$ capacitors :-



- (1) 1 : 2 (2) 4 : 5 (3) 1 : 5 (4) 1 : 4

18. Find the equivalent capacitance between A and B. (all capacitors are in μF)



- (1) $32 \mu\text{F}$ (2) $48 \mu\text{F}$ (3) $8 \mu\text{F}$ (4) $4 \mu\text{F}$

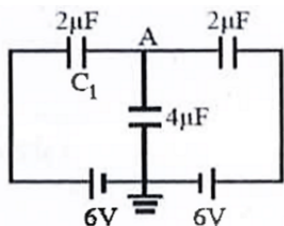
19. A capacitance of $2 \mu\text{F}$ is required in an electrical circuit across a potential difference of 1.0 kV . A large number of $1 \mu\text{F}$ capacitors are available which can withstand a potential difference of not more than 300 V .

The minimum number of capacitors required to achieve this is :

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- (1) 24 (2) 32 (3) 2 (4) 16

20. Three capacitors are connected as shown in fig. Then the charge on capacitor C_1 is :-

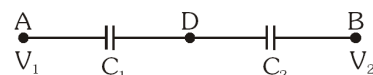


- (1) $6 \mu\text{C}$ (2) $12 \mu\text{C}$ (3) $18 \mu\text{C}$ (4) $24 \mu\text{C}$

21. Capacity of the capacitor of plates area A_1 and A_2 ($A_2 < A_1$) placed in front of each other is given by (plate separation is d) :-

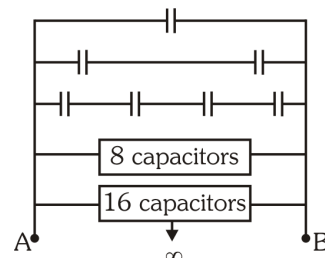
- (1) $\frac{\epsilon_0 A_1}{d}$ (2) $\frac{\epsilon_0 A_2}{d}$
(3) $\frac{\epsilon_0 (A_1 - A_2)}{d}$ (4) $\frac{\epsilon_0 (A_1 + A_2)}{2d}$

22. Two capacitances C_1 and C_2 in a circuit are joined as shown in figure. The potential of point A is V_1 and that of B is V_2 . The potential of point D will be :



- (1) $\frac{1}{2} (V_1 + V_2)$ (2) $\frac{C_2 V_1 + C_1 V_2}{C_1 + C_2}$
(3) $\frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$ (4) $\frac{C_2 V_1 - C_1 V_2}{C_1 + C_2}$

23. An infinite number of identical capacitors each of capacitance $1 \mu\text{F}$ are connected as in the adjoining figure. Then the equivalent capacitance between A and B is :



- (1) $1 \mu\text{F}$ (2) $2 \mu\text{F}$ (3) $\frac{1}{2} \mu\text{F}$ (4) ∞

24. Two coherent light beams of intensity I and $4I$ are superposed. The maximum and minimum possible intensities in the resulting beam are :

- (1) $5I$ and $3I$ (2) $5I$ and I
(3) $9I$ and $3I$ (4) $9I$ and I

25. The fringe width in Young's double slit experiment on a screen which is placed at a distance of 1 m from the slits 10^{-3} m apart when the light used has wavelength $6 \times 10^{-7} \text{ m}$ is equal to :-

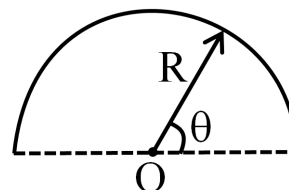
- (1) $3 \times 10^{-10} \text{ m}$ (2) $3 \times 10^{-4} \text{ m}$
(3) $6 \times 10^{-10} \text{ m}$ (4) $6 \times 10^{-4} \text{ m}$

26. A polaroid is placed at 45° to an incoming light of intensity I_0 . Now the intensity of light passing through the polaroid after polarisation would be -
 (1) I_0 (2) $I_0/2$ (3) $I_0/4$ (4) zero
27. The diameter of objective of a telescope is 1 m. Its resolving limit for wave length $4500 \text{ \AA}$, will be :-
 (1) $5.49 \times 10^{-7} \text{ rad}$ (2) $2.54 \times 10^{-4} \text{ rad}$
 (3) $6.54 \times 10^{-7} \text{ rad}$ (4) None of these
28. If the distance between the first maxima and fifth minima in YDSE is 7mm and the slit are separated by 0.15mm with the screen 50cm from the slits then wavelength of light used is :-
 (1) 420 nm (2) 467 nm
 (3) 525 nm (4) 600 nm
29. n incoherent sources each emitting beams of intensity I_0 are superimposed at a point, the resultant intensity of the point is :-
 (1) nI_0 (2) $\frac{I_0}{n}$
 (3) $n^2 I_0$ (4) None
30. Light of wavelength 589.3 nm is incident normally on the slit of width 0.1mm. What will be the angular width of the central diffraction maximum at a distance of 1m from the slit :-
 (1) 0.0117 radian (2) 0.53 radian
 (3) 0.0459 radian (4) 0.0314 radian
31. A beam of light of wavelength 600 nm from a distant source falls on a single slit 1 mm wide and the resulting diffraction pattern is observed on a screen 2m away. The distance between the first dark fringes on either side of the central bright fringe is :-
 (1) 1.2 mm (2) 1.2 cm
 (3) 2.4 cm (4) 2.4 mm
32. Ratio of intensities of two light waves is given by 4 : 1. The ratio of the amplitudes of two waves is :-
 (1) 2 : 1 (2) 1 : 2 (3) 4 : 1 (4) 1 : 4
33. YDSE is performed in air and then performed in water, the fringe width :-
 (1) Will remain same
 (2) Will decrease
 (3) Will increase
 (4) Depends on the type of wave.
34. In YDSE, for light of which colour, the fringe width will be minimum :-
 (1) Violet (2) Red
 (3) Green (4) Yellow
35. 'Electromagnetic waves are transverse in nature' is evident by :-
 (1) Polarization (2) Interference
 (3) Reflection (4) Diffraction

SECTION-B (PHYSICS)

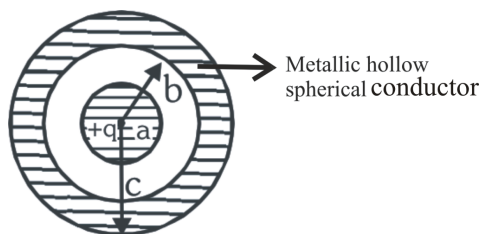
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36. The density of a material in SI units is 128 kg/m^3 . In certain unit in which the unit of length is 25 cm and the unit of mass is 50g, the numerical value of density of the material is :
 (1) 40 (2) 640 (3) 16 (4) 410
37. Linear charge density of semicircular arc varies with θ as $\lambda = \lambda_0 \sin \theta$ where λ_0 is a constant, then total charge on arc is :-



- (1) $2\lambda_0 R$ (2) $\lambda_0 R$
 (3) $\frac{2\lambda_0 R}{\pi}$ (4) $\frac{\lambda_0 R}{\pi}$

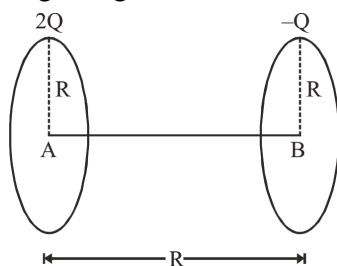
38. A solid non conducting sphere of radius 'a' has a net charge '+q' uniformly distributed throughout its volume. A hollow spherical conductor having inner and outer, radii 'b' and 'c' and net charge '-q' is concentric with the sphere (see the figure) Read the following statements.



- (i) The electric field at a distance r from the centre of the sphere $= \frac{1}{4\pi\epsilon_0} \frac{qr}{a^3}$ for $r < a$.
 (ii) The electric field at distance r from the centre of the sphere for $a < r < b = 0$
 (iii) The electric field at distance r from the centre of the sphere for $b < r < c = 0$
 (iv) The charge on the inner surface of the hollow sphere $= -q$
 (v) The charge on the outer surface of the hollow sphere $= +q$

Which of the above statements are true ?

- (1) (i), (ii) and (v) (2) (i), (iii) and (iv)
 (3) (ii), (iii) and (iv) (4) (ii), (iii) and (v)
39. In the following arrangement. Choose correct option:-



- (1) $V_A = \frac{Q(2\sqrt{2} - 1)}{4\pi\epsilon_0\sqrt{2}R}$
 (2) $V_B = \frac{Q(\sqrt{2} - 1)}{4\pi\epsilon_0 R}$

W.D. in moving a charge $+q$ from B to A

- (3) $W_{BA} = \frac{qQ(3\sqrt{2} - 3)}{4\sqrt{2}\pi\epsilon_0 R}$
 (4) all options are correct

40. A $20\mu\text{F}$ capacitor in a defibrillator is charged to 2000 V. The energy stored in the capacitor is sent through the patient during a pulse of duration 4ms. The power delivered to the patient is :-

- (1) 10 kW
 (2) 20 kW
 (3) 30 kW
 (4) 40 kW

41. A parallel plate air capacitor has a capacitance C . When it is half filled with a dielectric of dielectric constant 5, the percentage increase in the capacitance will be :-



- (1) 400%
 (2) 66.6%
 (3) 33.3%
 (4) 200%

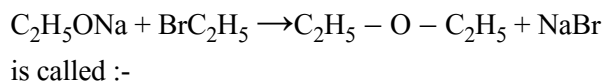
42. A parallel plate capacitor is connected with a battery whose potential difference remains constant. If the plates of the capacitor are shifted apart then the intensity of electric field :

- (1) decreases and charge on plates also decreases.
 (2) remains constant but charge on plates decreases.
 (3) remains constant but charge on the plates increases.
 (4) increases but charge on the plates decreases.

Topic : NAR, ESR, FRSR, NSR, SOLID STATE, SURFACE CHEMISTRY, PERIODIC TABLE [BACK UNIT] :
SOLUTION TG: @Chalnaayaaar

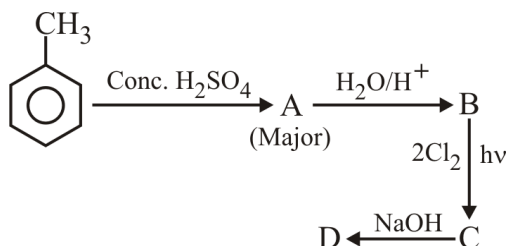
SECTION-A (CHEMISTRY)

51. The reaction



- (1) Frankland reaction
- (2) Wurtz reaction
- (3) Williamson's synthesis
- (4) Cannizzaro reaction

52.



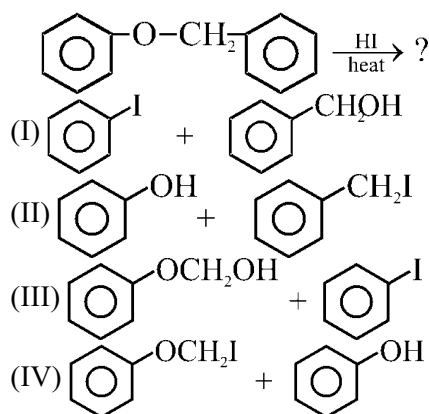
Correct statement ?

- (1) B is more reactive towards ESR than toluene
- (2) C is O-chloro toluene
- (3) When D reacts with water, it forms stable hydrate
- (4) D forms racemic mixture on adding HCN

53. Salicylic acid can be easily prepared by reaction between :-

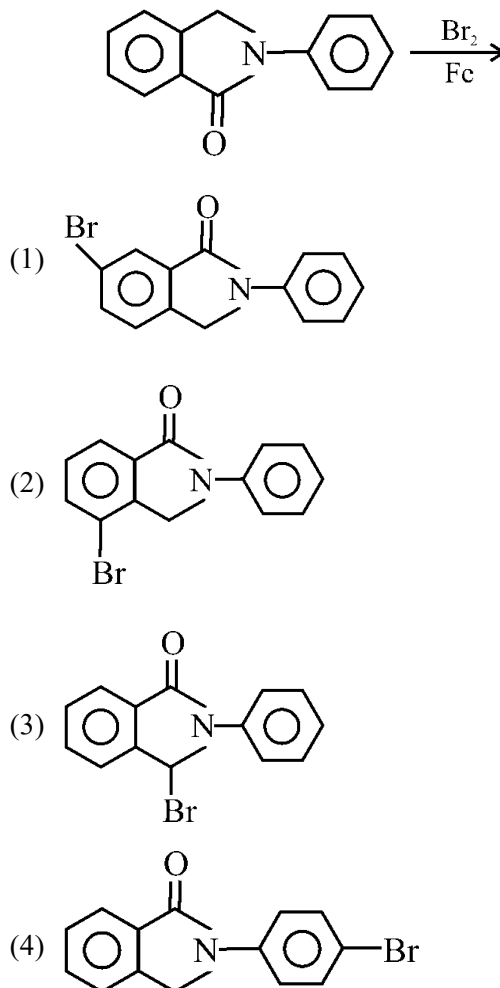
- (1) Phenol and CO_2/NaOH
- (2) Benzoic acid and H_2O_2
- (3) Benzene diazonium chloride and CO_2
- (4) Phenol and formic acid

54. What are the major products from the following reaction ?



- (1) I (2) II (3) III (4) IV

55. In the reaction the major product formed is :-



56. When ethyl chloride and n-propyl chloride undergoes wurtz reaction which is not obtained :-

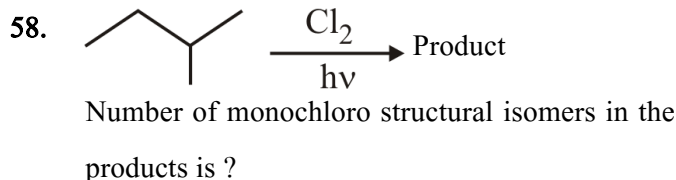
- (1) n-butane
- (2) n-pentane
- (3) n-Hexane
- (4) Isobutane

57. Which amongst the following reactions of alkyl Halide produces isonitrile as a major product ?

- (1) $\text{R-X} + \text{HCN} \rightarrow$
- (2) $\text{R-X} + \text{AgCN} \rightarrow$
- (3) $\text{R-X} + \text{KCN} \rightarrow$
- (4) $\text{R-X} + \text{NaCN} \xrightarrow{\text{H}_2\text{O}}$

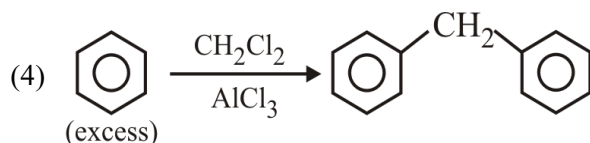
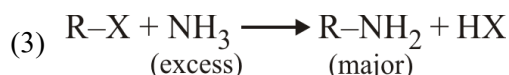
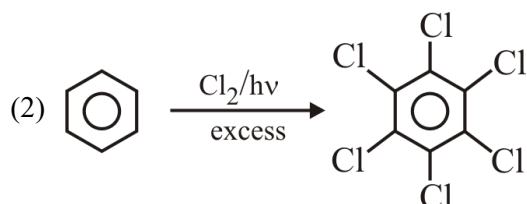
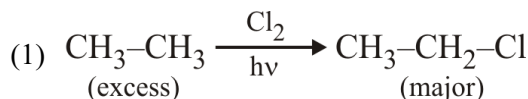
English

9



- (1) 3 (2) 4
(3) 6 (4) 5

59. In which of the following reaction, product formed is wrong :-

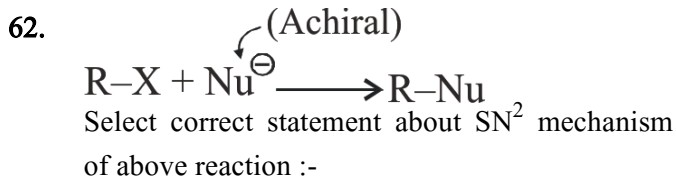


60. Which is correct nucleophilic strength of following nucleophile :-

- (1) $\text{OH}^- > \text{CH}_3\text{COO}^- > \text{OCH}_3^- > \text{C}_6\text{H}_5\text{O}^-$
(2) $\text{CH}_3\text{O}^- > \text{CH}_3\text{COO}^- > \text{C}_6\text{H}_5\text{O}^- > \text{OH}^-$
(3) $\text{C}_6\text{H}_5\text{O}^- > \text{OH}^- > \text{CH}_3\text{O}^- > \text{CH}_3\text{COO}^-$
(4) $\text{CH}_3\text{O}^- > \text{OH}^- > \text{C}_6\text{H}_5\text{O}^- > \text{CH}_3\text{COO}^-$

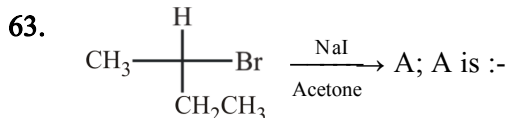
61. Which of following reaction does not involves carbocation intermediate ?

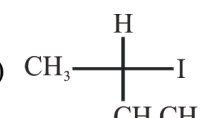
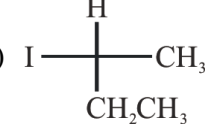
- (1) Addition of HBr at alkene
(2) SN^1 reaction at alkyl halide
(3) Friedel craft alkylation
(4) Halogenation of alkane



- (1) If R-X is optically active then product is racemic mixture
(2) If R-X is optically inactive then product has inversion of configuration
(3) If R-X is optically active then product has retention of configuration
(4) If R-X is optically active then product has inversion of configuration

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- (1) 
(2) 
(3) 
(4) Both (1) & (3)

64. Match the column I with column II and choose the correct option using the codes given below

	Column-I (Number of periods)		Column II (Number of elements)
(A)	First period	(1)	14
(B)	Third period	(2)	2
(C)	Lanthanoids	(3)	8
(D)	Actinoids	(4)	4

- (1) A-2, B-4, C-1, D-3
(2) A-2, B-3, C-1, D-1
(3) A-4, B-2, C-1, D-3
(4) A-4, B-2, C-3, D-1

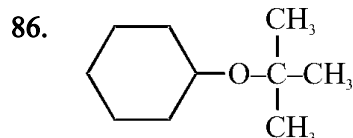
65. Correct electronic configuration of atomic number 115 is-
- $[\text{Xe}] 6s^2, 4f^{14}, 5d^{10}, 6p^3$
 - $[\text{Rn}] 7s^2, 5f^{14}, 6d^{10}, 7p^3$
 - $[\text{Rn}] 7s^2, 5f^{14}, 6d^{10}, 7p^1$
 - $[\text{Xe}] 6s^2, 4f^{14}, 5d^{10}, 6p^2$
66. If aufbau rule is not followed then Ca belongs to which block ?
- s-block
 - d-block
 - p-block
 - f-block
67. Elements with atomic number 10, 18, 36, 54 and 86 are all :-
- TG: @Chalnaayaaar
- light metals
 - inert gases
 - halogens
 - rare earth
68. What will be the name of next discovered alkaline earth metal in 8th period :-
- Uub
 - Uuo
 - Ubn
 - Ubu
69. Match the column:
- | | |
|---------------|-------------|
| (A) $Z = 102$ | (P) s-block |
| (B) $Z = 120$ | (Q) d-block |
| (C) $Z = 107$ | (R) f-block |
| (D) $Z = 82$ | (S) p-block |
- A-R, B-P, C-Q, D-S
 - A-Q, B-P, C-S, D-R
 - A-R, B-P, C-S, D-Q
 - A-P, B-Q, C-R, D-S
70. Which of the following element does not belong to III B group :-
- $Z = 72$
 - $Z = 63$
 - $Z = 57$
 - $Z = 89$
71. **Assertion** : Molarity of a solution in liquid state changes with temperature.
Reason : The volume of a solution changes with change in temperature.
- Assertion is correct, reason is correct; reason is a correct explanation for assertion.
 - Assertion is correct, reason is correct; reason is not a correct explanation for assertion
 - Assertion is correct, reason is incorrect
 - Assertion is incorrect, reason is correct.
72. The van't Hoff factor i for an electrolyte which undergoes dissociation and association in solvent are respectively :-
- Greater than one and less than one
 - Less than one and greater than one
 - Less than one and less than one
 - Greater than one and greater than one
73. If liquid A and B form ideal solution, then:
- $\Delta G_{\text{mix}} = 0$
 - $\Delta H_{\text{mixing}} = 0$
 - $\Delta G_{\text{mix}} = 0, \Delta S_{\text{mix}} = 0$
 - $\Delta S_{\text{mix}} = 0$
74. Identify the gas which is readily adsorbed by activated charcoal :-
- N_2
 - SO_2
 - H_2
 - O_2

75. Which of the following affects the adsorption of gas on the surface of solid ?
- Critical temperature
 - Pressure of gas
 - Temperature
 - All of these
76. A catalyst in the finely divided form is most effective because :-
- less surface area is available
 - more active sites are formed
 - more energy gets stored in the catalyst
 - none of these
77. A compound is formed by elements A and B. This crystallises in the cubic structure when atoms A are at the corners of the cube and atoms B are at the centre of the body. The simplest formula of the compound is:
- AB
 - AB₂
 - A₂B
 - AB₄
78. Milk is an example of colloid :-
- Liquid - liquid
 - Solid - liquid
 - Liquid - solid
 - Liquid - gas
79. Which of the following show negative deviation from ideal behaviour :-
- C₆H₆ + C₆H₅-CH₃
 - CHCl₃ + CH₃COCH₃
 - C₂H₅Cl + C₂H₅Br
 - C₂H₅OH + H₂O
80. Value of Henry constant for the gases A, B, C and D are 10⁻² atm⁻¹, 10⁻³ atm⁻¹, 2 × 10⁻³ atm⁻¹ and 10⁻⁴ atm⁻¹. Which of the following gas has maximum solubility?
- D
 - C
 - B
 - A
81. Which of the following solution will have highest osmotic pressure at same temperature?
- 1 m NaCl(aq) solution
 - 1 m MgCl₂(aq) solution
 - 1 m Hg₂Cl₂(aq) solution
 - 4 m C₆H₁₂O₆(aq) solution
82. Mole fraction of A vapours above solution in mixture of A and B ($X_A = 0.4$, $P_A^0 = 100$ mm, $P_B^0 = 200$ mm) will be :
- 0.4
 - 0.8
 - 0.25
 - None of these
83. The Van't Hoff factor (i) for an aqueous solution of K₃[Fe(CN)₆] is if ($\alpha = 90\%$)
- 4
 - 3.7
 - 4.3
 - $\frac{1}{4}$
84. The molarity (M) of pure water is :-
- 1M
 - 10⁻⁷M
 - 55.5M
 - 18M

85. Which of the following solution has highest normality (N) :-

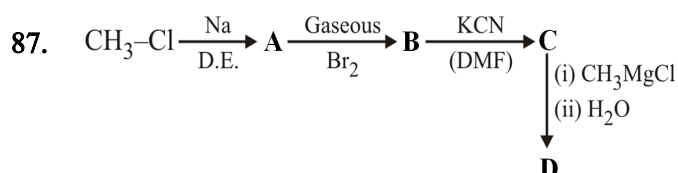
- (1) 1 M HCl(aq) solution
- (2) 1 M H₂SO₄(aq) solution
- (3) 1 M H₃PO₄(aq) solution
- (4) All have same normality

SECTION-B (CHEMISTRY)



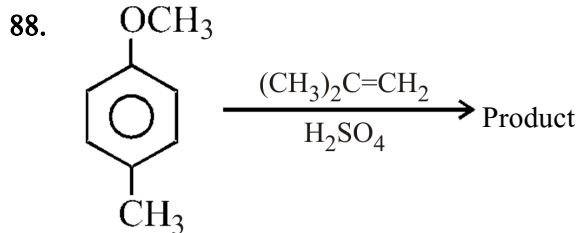
Above ether is best prepared by :-

- (1)
- (2)
- (3)
- (4)



Structure of 'D' is ?

- (1)
- (2)
- (3)
- (4)



product is :-

- (1)
- (2)
- (3)
- (4)

89. $\text{CH}_3\text{-CH}_2\text{-COONa} \xrightarrow{\text{K.E.}}$ incorrect statement ?

- (1) Product formed is butane
- (2) pH of reaction increases
- (3) CO₂ is liberated at anode
- (4) Product formed is ethane

90. Acetophenone reacts with semicarbazide product will be :-

- (1)
- (2)
- (3)
- (4)

91. Identify d block element :

- (1) $[\text{Xe}] 4f^{14} 5d^{10} 6s^2$
- (2) $[\text{Xe}] 4f^{14} 5d^1 6s^2$
- (3) $[\text{Rn}] 5f^{14} 6d^1 7s^2$
- (4) All of above

92.

	Column-I (Type of Elements)		Column-II (Outer electronic configuration)
(A)	Inert gas elements	(I)	ns^{1-2} and $ns^2 np^{1-5}$
(B)	Representative elements	(II)	$1s^2$ and $ns^2 np^6$
(C)	Transition elements	(III)	$(n-2)f^{1-14}$ $(n-1)d^{0-1} ns^2$
(D)	Inner transition elements	(IV)	$(n-1)d^{1-10} ns^{0-2}$

- (1) A-i, B-ii, C-iii, D-iv
- (2) A-ii, B-i, C-iii, D-iv
- (3) A-ii, B-i, C-iv, D-iii
- (4) A-i, B-ii, C-iv, D-iii

93. Period number and group number of element ($z = 73$) are respectively :-

- (1) 5, 7
- (2) 6, 13
- (3) 6, 5
- (4) 5, 6

94. Colligative properties depend on the :-

- (1) Relative no. of solute molecules in solution and the nature of the solvent
- (2) Relative no. of solute molecules in solvent and the nature of solute
- (3) Relative no. of solute molecules and the nature of solute and solvent
- (4) Relative no. of solute molecules, irrespective of the nature of solvent and solute

95. Van't Hoff factor is given by the expression _____.

- (1) $i = \frac{\text{Normal molar mass}}{\text{Abnormal molar mass}}$
- (2) $i = \frac{\text{Abnormal molar mass}}{\text{Normal molar mass}}$
- (3) $i = \frac{\text{Observed colligative property}}{\text{Calculated colligative property}}$
- (4) Both (1) and (3)

96. the substance A when dissolved in solvent B shows the molecular mass corresponding to A_3 . the van't Hoff factor will be

- (1) 1
- (2) 2
- (3) 3
- (4) $1/3$

97. At higher altitudes the boiling point of water lowers because.

- (1) Atmospheric pressure is low
- (2) Temperature is low
- (3) Atmospheric pressure is high
- (4) None of these

98. What weight of solute (molecular weight = 60) is required to dissolve in 180 g of water to reduce the vapour pressure to $\frac{4}{5}$ th of pure water ?

- (1) 48 g
- (2) 96 g
- (3) 150 g
- (4) 175 g

99. The number of atoms per unit cell in a simple cubic, fcc and bcc structure respectively are :-

- (1) 1, 4, 2
- (2) 1, 2, 4
- (3) 8, 14, 9
- (4) 8, 4, 2

100. Frenkel defect is found in crystal in which :-

- (1) radius ratio is low
- (2) coordination no is high
- (3) coordination no is low
- (4) both (1) & (3)

Topic : MOLECULAR BASIS OF INHERITANCE (TILL TAUGHT), ECOLOGY : NEGATIVE INTERACTIONS TO ECOSYSTEM [BACK UNIT] : ORIGIN EVOLUTION

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SECTION-A (BIOLOGY-I)

101. In a food chain, the total amount of living material, is shown by :-

- (1) Pyramid of energy
- (2) Pyramid of numbers
- (3) Pyramid of biomass
- (4) Both (1) & (3)

102. Warm blooded animals (birds and mammals) have larger body size in cold climate than in hotter areas, is :-

- (1) Allen's Rule
- (2) Jordan's Rule
- (3) Rensch's Rule
- (4) Bergman's Rule

103. Which of the following is not a limitation of pyramids :-

- (1) It does not take into account the same species belonging to two or more trophic level
- (2) It does not accommodate a food chain
- (3) It does not accommodate a food web
- (4) Saprophytes are not given any place

104. Which of the following is slowest step of decomposition ?

- (1) Fragmentation
- (2) Mineralisation
- (3) Humification
- (4) Leaching

105. Match the column :-

	I		II
A	Fragmentation	i	Leads to accumulation of a dark coloured substance
B	Humification	ii	Water-soluble inorganic nutrients undergo in soil horizon
C	Catabolism	iii	Bacterial and fungal enzymes degrade detritus into simple organic and inorganic substance
D	Leaching	iv	Detritivores break down detritus in small pieces

- (1) a-iii, b-i, c-iv, d-ii
- (2) a-iv, b-i, c-ii, d-iii
- (3) a-iv, b-iii, c-i, d-ii
- (4) a-iv, b-i, c-iii, d-ii

106. Some nutrients gets attached to the biomass of microbes which becomes unavailable to other organisms, it is known as :-

- (1) Acclimatization
- (2) Leaching
- (3) Mineralization
- (4) Nutrient immobilization

107. Solar energy converted to chemical energy by plant and chemical energy get converted into kinetic or heat energy in animals, it shows :-

- (1) 1st law of thermodynamic in food chain
- (2) 2nd law of thermodynamic in food chain
- (3) Above both laws in food chain
- (4) Above statement is not related with law of thermodynamic

- 108.** Which is not the example of link between two ecosystems ?
- A leaf of tree falling into pond
 - A fish is eaten by bird
 - A fish is eaten by man
 - Grass is eaten by goat
- a, b, c
 - b & d
 - c & d
 - Only d
- 109.** Read statements regarding population interaction:
- Biological control methods are adopted in agricultural pest control based on predation
 - In the rocky intertidal areas of American pacific coast starfish is an important predator.
 - Some species of frog & insect show camouflage
 - Host specific Parasite and host show co-evolution
- How many of above statements are **correct** ?
- Statement A, B, C, D
 - Only statement D
 - Only statement C, D
 - Only A, B and D
- 110.** Ecological niche can be best defined as ?
- Immediate physical area of organism
 - Reproduction ability
 - Sum total of biotic components in a ecosystem
 - Functional role of a species in a ecosystem
- 111.** Which of following most adversely affected by competition ?
- Herbivores
 - Carnivores
 - Plants
 - Apex predators
- 112. Statement-I :** But for predators, prey populations could achieve high numbers.
- Statement-II :** Predators are too efficient for predation.
- Statement-I & Statement-II both are correct.
 - Statement-I is correct & Statement-II is incorrect.
 - Statement-I is incorrect & Statement-II is correct.
 - Both Statement-I & Statement-II is incorrect.
- 113. Statement-I :** Competition can occur only in closely related species, feeding on limited resources.
- Statement-II :** Two species cannot occupy same niche.
- Statement-I & Statement-II both are correct.
 - Statement-I is correct & Statement-II is incorrect.
 - Statement-I is incorrect & Statement-II is correct.
 - Both Statement-I & Statement-II is incorrect.
- 114. Assertion (A) :** Ecosystem follows both laws of thermodynamics.
- Reason (R) :** The ultimate source of energy is the sun for all ecosystem.
- Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 - Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 - Assertion is True but Reason is False
 - Both Assertion & Reason are False.

115. Select the correct statement :

- (1) Trophic level represents species
- (2) Two species cannot occupy same trophic level
- (3) Single species can occupy more than one trophic level at same time
- (4) Primary consumers are placed at first trophic level

116. Competitive exclusion principle is given by :

- (1) Connell
- (2) Mac Arthur
- (3) Gause
- (4) Odum

117. Which of the following is an example of man-made ecosystem ?

- (1) Estuary
- (2) Sea
- (3) Aquarium
- (4) Forest

118. Organisms with wide range of thermal tolerance are known :

- (1) Euryhaline
- (2) Eurythermal
- (3) Stenothermal
- (4) Stenohaline

119. All birds & mammals are :

- (1) Regulators
- (2) Conformers
- (3) Partial conformer
- (4) Partial regulators

120. According to Robert Constanza's estimate how much percentage of total cost is for soil formation ?

- (1) 20%
- (2) 30%
- (3) 50%
- (4) 70%

121. NPP of whole biosphere annually is :

- (1) 4×10^{13} kg
- (2) 170 billion ton
- (3) 115 billion ton
- (4) 55 billion ton

122. How much carbon is fixed annually in the biosphere through photosynthesis ?

- (1) 13×10^4 kg
- (2) 4×10^{13} ton
- (3) 170 billion ton
- (4) 4×10^{13} kg

123. Which of the following process is antagonistic to nutrient immobilisation ?

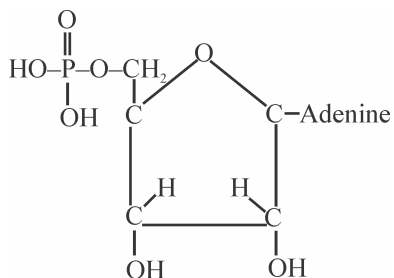
- (1) Fragmentation
- (2) Catabolism
- (3) Mineralization
- (4) Leaching

124. The main reservoir for phosphorus cycle is :

- (1) Rocks
- (2) Ocean
- (3) Atmosphere
- (4) Limestone

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125. Following structure is :-



- (1) Deoxyribonucleotide
- (2) Ribonucleotide
- (3) Deoxyribonucleoside
- (4) Ribonucleoside

126. Which of the following is correct for Watson and Cricks model of DNA. It is double stranded helix with :

- (1) 20 base pair and 3.4 nm distance between two successive steps
- (2) 20 base pair and 3.4 Å distance between two base pair
- (3) 10 base pair and 3.4 nm distance between two base pair
- (4) 10 base pair and 3.4 Å distance between two successive steps

127. Match list-I with list-II :-

	List-I		List-II
(A)	Escherichia coli	(I)	3.3×10^9 bp
(B)	Human DNA (Haploid)	(II)	6.6×10^9 bp
(C)	$\phi \times 174$ bacteriophage	(III)	5386 nucleotide
(D)	Lambda bacteriophage	(IV)	4.6×10^6 bp
		(V)	48502 bp

Choose the correct answer from the options given below :-

- (1) A-I, B-III, C-II, D-IV
- (2) A-IV, B-I, C-III, D-V
- (3) A-III, B-II, C-IV, D-I
- (4) A-IV, B-II, C-III, D-V

128. Given below are two statements :

Statement-I :- The negatively charged DNA is wrapped around the positively charged histone octamer to form a structure called nucleoid.

Statement-II :- The beads-on-string structure in chromatin is packaged to form chromatin fibers that are further coiled and condensed at metaphase stage of cell division to form chromosomes.

In the light of the above statements, choose the most appropriate answer from the options given below :-

- (1) Both statement-I and statement-II are incorrect
- (2) Statement-I is correct but statement-II is incorrect
- (3) Statement-I is incorrect but statement-II is correct
- (4) Both statement-I and statement-II are correct

129. **Assertion** : Euchromatin is said to be transcriptionally active chromatin.

Reason : Due to loose coiling of euchromatin RNA polymerase can easily move up on it.

- (1) Both Assertion & Reason are true & the Reason is a correct explanation of Assertion
- (2) Both Assertion & Reason are true & the Reason is not a correct explanation of Assertion
- (3) Assertion is true but the Reason is False.
- (4) Both Assertion & Reason are False

130. Avery, Macleod and McCarty worked to determine the biochemical nature of transforming principle in Griffith's experiment. The process of transformation is not affected by which of the following enzymes ?

- (A) DNase (B) RNAase
(C) Lipase (D) Proteases

- (1) A,C,D (2) A,B,D
- (3) D,A,C (4) B,D,C

131. Assertion : Among the two nucleic acids, the DNA is a better genetic material.

Reason : DNA chemically is less reactive and structurally more stable when compared to RNA.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

132. A normal DNA molecule is continuously replicated in N^{15} medium then what is the % of non-radioactive DNA in 4th generation :

- (1) 12.5% (2) 25% (3) 0% (4) 6.25%

133. Fill in the blanks :

(A) It is the presence of ___a___ in a transcription unit that defines the ___b___ and coding strands.

(B) The terminator is located to the ___c___ and the promoter is located to the ___d___ of the structural gene.

- (1) a-terminator, b-template, c-upstream, d-downstream
- (2) a-promoter, b-template, c-downstream, d-upstream
- (3) a-promoter, b-downstream, c-template, d-upstream
- (4) a-template, b-downstream, c-upstream, d-terminator

134. 3' A A A A A A A 5' Template strand

5' T T T T T T T 3' Coding strand

What is the sequence of RNA molecule which is transcribed by above hypothetical DNA sequence.

- (1) 5' T T T T T T T 3'
- (2) 5' A A A A A A A 3'
- (3) 5' U U U U U U U 3'
- (4) 3' T T T T T T T 5'

135. Assertion :- The presence of introns is reminiscent of antiquity and the process of splicing represents the dominance of RNA world.

Reason :- The split gene arrangement represents an advanced feature of the genome.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

SECTION-B (BIOLOGY-I)

136. Producers $\longrightarrow$ Consumers

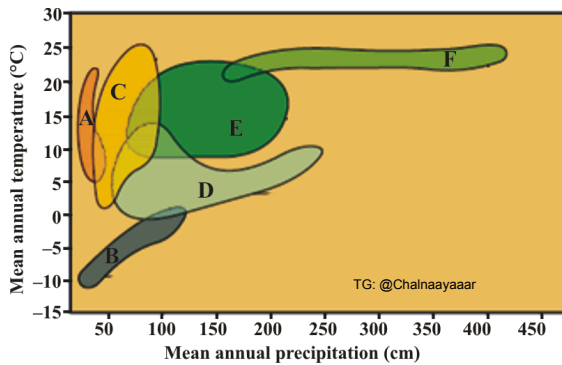
↓
DFC

↓
DFC

Which of the following is incorrect and irrelevant for the line diagram given above.

- (1) Every living member (producer, consumer and decomposer) need a constant supply of nutrients to grow, reproduce and regulate various body functions.
- (2) Nutrients are never lost from the ecosystems, they are recycled again and again indefinitely.
- (3) This diagram represents cyclic flow of energy
- (4) When any organism dies it is converted to detritus or dead biomass that serves as an energy source for decomposers

137. Which of the following statement is correct with respect to different biomes represented in the given figure :-



- (1) Biome 'A' and 'F' show high annual temperature and they are highly rich in vegetation
- (2) Biome 'C' and 'D' differ in temperature as well as rainfall but their vegetation type is similar
- (3) Biome 'A' and 'B' differ in their mean annual temperature but both have very low vegetation
- (4) Biome 'A' and 'C' have low precipitation and thus vegetation type is similar
138. How many statements are correct ?
- (a) Net primary productivity is available biomass for consumption to heterotrophs
- (b) Secondary productivity is defined as rate of formation of new organic matter by heterotrophs
- (c) The annual net primary productivity of ocean is greater than land
- (d) Warm and moist environment favours decomposition
- (1) abc and d (2) ab and c
- (3) ab and d (4) a and c only
139. Choose the correct match for food chain :-

		A	B	C
(i)	% of energy transfer	Increases	Decreases	Constant
(ii)	Amount of energy	Increases	Decreases	Constant
(iii)	Respiratory cost	Increases	Decreases	Constant
(iv)	Intensity of competition	Increases	Decreases	Constant

- (1) (i)-C (ii)-A (iii)-A (iv)-B
- (2) (i)-B (ii)-C (iii)-A (iv)-C
- (3) (i)-C (ii)-B (iii)-A (iv)-A
- (4) (i)-C (ii)-B (iii)-A (iv)-B

140. Which one is not a adaptation of altitude sickness ?

- (1) Increasing red blood cell production
- (2) Decreasing the binding capacity of haemoglobin with oxygen
- (3) Increasing breathing rate
- (4) Increase in heart palpitation

141. Read the following statements :

- (i) Totally unrelated species can compete for the same resource
- (ii) Resources need not always be limiting for competition to occur
- (iii) In 'Interference competition', the feeding efficiency of one species might be reduced due to the interfering and inhibitory presence of the other species
- (iv) Abingdon tortoise in Galapagos islands became extinct within a decade after goats were introduced, apparently due to the greater browsing efficiency of goats
- How many of the above statements are correct ?

- (1) Four (2) One
- (3) Two (4) Three

142. Antibiotics production is based on :

- (1) Predation (2) Mutualism
- (3) Amensalism (4) Parasitism

143. Match the column-I with column-II.

	Column-I		Column-II
(A)	Lion	(i)	Producers
(B)	Plants	(ii)	Secondary consumer
(C)	Deer	(iii)	Top carnivore
(D)	Fox	(iv)	Primary consumer

- (1) A-iii, B-i, C-iv, D-ii
- (2) A-iii, B-ii, C-iv, D-i
- (3) A-iv, B-i, C-iii, D-ii
- (4) A-iv, B-iii, C-ii, D-i

144. Which of the following is not a symptom of altitude sickness ?

- (1) High breathing rate
- (2) Nausea
- (3) Fatigue
- (4) Heart palpitation

145. Pyramid of biomass in aquatic ecosystem :

- (1) Always upright
- (2) Upright in pond
- (3) Inverted
- (4) Always spindle

146. **Assertion (A)** : Pyramid of energy is always upright.

Reason (R) : Ecosystem follows second law of thermodynamics.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

147. Which one of the following DNA will have the highest melting temperature?

- (1) DNA with 50% Adenine
- (2) DNA with 45% Thymine
- (3) DNA with 10% Guanine
- (4) DNA with 35% Cytosine

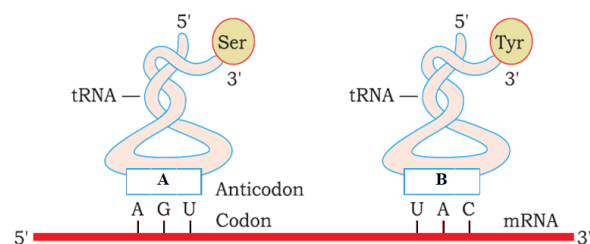
148. The following diagram refers to Griffin's demonstration of transformation in *Pneumococcus*. A, B, C and D indicate the fates of mice after they are injected with specific bacteria. Identify the fate of mice.

S strain → Inject into mice → ...A...
 R strain → Inject into mice → ...B...
 S strain → Injected into mice → ...C...
 (heat killed)
 S strain → Inject into mice → ...D...
 (heat killed) + R strain

- (1) A-dies, B-dies, C-lives, D-lives
- (2) A-lives, B-lives, C-dies, D-dies
- (3) A-dies, B-lives, C-lives, D-dies
- (4) A-lives, B-dies, C-dies, D-lives

149. Following is a diagrammatic presentation of the adapter molecule.

What will the correct sequence of anticodons for serine and tyrosine tRNA respectively?



- (1) A – 5' UCA 3' B – 5' AUG 3'
- (2) A – 5' AUG 3' B – 5' UCA 3'
- (3) A – 5' ACU 3' B – 5' GUA 3'
- (4) A – 5' GUA 3' B – 5' ACU 3'

150. Which of the following process does not operate in template independent manner?

- (1) Addition of adenylate residues in tailing
- (2) Enzymatic synthesis of RNA by severo ocha enzyme in-vitro
- (3) Copying of genetic information from DNA to RNA in transcription
- (4) All of the above

Topic : MOLECULAR BASIS OF INHERITANCE (TILL TAUGHT), ECOLOGY : NEGATIVE INTERACTIONS TO ECOSYSTEM [BACK UNIT] : ORIGIN EVOLUTION TG: @Chalnaayaaar

SECTION-A (BIOLOGY-II)

151. Which of the following is exclusive work of RNA polymerase core enzyme ?

- (1) Elongation (2) initiation
(3) termination (4) Splicing

152. Extinction of dinosaurs occurred in :-

- (1) Eocene and oligocene epochs
(2) Silurian and devonian periods
(3) Cretaceous period
(4) Jurassic and triassic period

153. Which of the following statement is incorrect :-

(1)	Theory of special creation	Diversity was same and will be same in future.
(2)	Theory of Biogenesis	New life can be originated on earth by decaying and rotting matter.
(3)	Cosmic panspermia theory	Life came from outer space through spores.
(4)	Oparin-Haldane theory	First life have come from pre existing non-living organic molecule

154. Which of the following statement is not true ?

- (1) Wings of Birds and Butterfly is an example of homology
(2) Forelimbs of mammals indicate common ancestry
(3) Flippers of penguins and dolphins show analogy
(4) Analogy indicates different ancestry

155. "Embryos of advanced species pass through stages represented by adult organism of more primitive species" the given statement is related with:-

- (1) Baer's law
(2) Biogenetic law
(3) Recapitulation theory
(4) Dollo's law

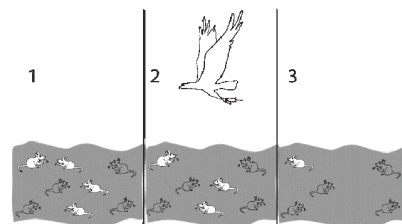
156. **Assertion (A):** New life forms of microbes are appearing in a time scale of months or years and not centuries.

Reason (R): The life span of microbes is less as compared to higher organisms.

In the light of above statements, choose the correct answer from the following options.

- (1) (A) is true but (R) is false
(2) (A) is false but (R) is true
(3) Both (A) and (R) are true and (R) is the correct explanation of (A)
(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

157.



The above picture shows that the white mouse is less fit. It is less likely to survive to reproductive age or produce many off spring due to the predator. Thus natural selection will select and favor the darker mouse which is more fit. Given example shows which type of natural selection ?

- (1) Directional selection
(2) Stabilizing selection
(3) Disruptive selection
(4) Artificial selection

158. Read the statements (i-iv) and choose the correct option

- i. Increase in melanised moths after industrialization in Great Britain is a proof of Natural Selection.
- ii. More individuals acquiring mean character value cause disruptive selection.
- iii. Change in allelic frequency leads to Hardy-Weinberg equilibrium.
- iv. Genetic drift changes allelic frequency in future generations.

- (1) ii is correct
- (2) i is correct
- (3) i and iv are correct
- (4) i and iii are correct

159. Which statement is false for Homo habilis ?

- (1) They were the first human like being
- (2) They probably did not eat meat
- (3) The brain capacity was between 650-800cc
- (4) They are included in ape-man fossil.

160. The chronological order of human evolution from early to the recent is :-

- (1) *Ramapithecus* → *Homo habilis* → *Australopithecus* → *Homo erectus*
- (2) *Australopithecus* → *Homo habilis* → *Ramapithecus* → *Homo erectus*
- (3) *Australopithecus* → *Ramapithecus* → *Homo habilis* → *Homo erectus*
- (4) *Ramapithecus* → *Australopithecus* → *Homo habilis* → *Homo erectus*

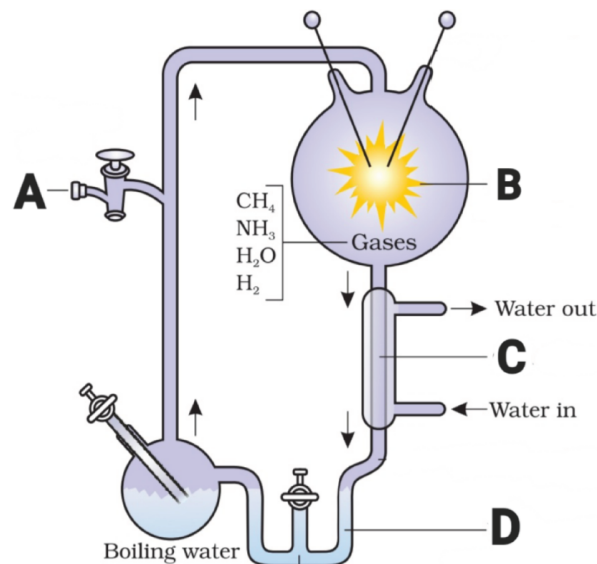
161. The origin of ____ is considered a unique event in the history of universe.

- (1) Life
- (2) Earth
- (3) Sun
- (4) Moon

162. Life appeared _____ years after the formation of earth.

- (1) 500 billion
- (2) 4 billion
- (3) 500 million
- (4) 4 million

163. Which part in the following setup gets connected to vacuum pump ?



- (1) D
- (2) C
- (3) B
- (4) A

164. **Assertion** : Those who are better fit in an environment, leave more progeny than others.

Reason : The fitness always refers only to reproductive fitness.

- (1) Both assertion and reason are true and reason is the correct explanation of assertion.
- (2) Assertion is true but reason is false
- (3) Both assertion and reason are false
- (4) Both assertion and reason are true and reason is not the correct explanation of assertion

165. Earnst Heckel's proposal was disapproved on careful study performed by :-

- (1) Karl Ernst von Baer
- (2) Muller
- (3) Alfred wallace
- (4) Hardy-Weinberg

166. How many of the following factors disturb Hardy-Weinberg equilibrium ?
random mating, gene recombination, gene migration, genetic drift, sexual selection, mutation.
- (1) Six (2) Five
(3) Four (4) Three
167. **Statement-I** : Change of frequency of alleles in a population be interpreted as resulting in evolution.
Statement-II : There would be a gene flow if gene migration, happens multiple times.
- (1) Both statement-I and statement-II are true
(2) Both statement-I and statement-II are false
(3) Statement-I is true but statement-II is false
(4) Statement-I is false but statement-II is true
168. Invertebrates were formed and active by the time of :-
- (1) 5 mya (2) 50 mya
(3) 500 mya (4) 5000 mya
169. With us, there are no specimens left of :-
- (1) Coelacanth (2) First amphibians
(3) Lobefins (4) Both (1) and (3)
170. The story of evolution of modern man is most interesting and appears to ____ evolution of human brain and language.
- (1) Convergent (2) Parallel
(3) Divergent (4) Both (1) and (2)
171. Which of the following shows adaptive convergence with Numbat ?
- (1) Anteater (2) Lemur
(3) Bandicoot (4) Koala
172. Direct common ancestor of most plants are :-
- (1) Tracheophytes (2) Chlorophytes
(3) Progymnosperms (4) Psilophytons
173. Which of the following has RNA but not DNA ?
- (1) Nucleus (2) Mitochondrion
(3) Chloroplast (4) Ribosome
174. Which of the following is not a nucleoside ?
- (1) Cytosine (2) Adenosine
(3) Guanosine (4) Uridine
175. How many phosphodiester bonds will be present in a prokaryote with 400bp DNA ?
- (1) 400 (2) 1598 (3) 800 (4) 798
176. Which of the following is not part of octamer ?
- (1) H1 (2) H2A (3) H3 (4) H4
177. Biochemical characterization of transforming principle was done by :-
- (1) Griffith (2) Hershey and Chase
(3) Avery (4) Macleod and McCarty
178. Performing Meselson and Stahl's experiment for 20 minutes will reject idea of which mode of DNA replication ?
- (1) Dispersive (2) Conservative
(3) Semiconservative (4) Both (1) and (2)
179. Which of the following is not required for unwinding of DNA helix ?
- (1) Helicase (2) SSB
(3) Topoisomerase (4) Primase
180. Which of the following is most stable ?
- (1) t-RNA (2) m-RNA
(3) r-RNA (4) hn-RNA
181. hn-RNA and m-RNA are synthesized by :-
- (1) RNA polymerase-I
(2) RNA polymerase-II
(3) RNA polymerase-III
(4) Primase

182. Which of the following has split gene arrangement ?

- (1) Eubacteria (2) Eukaryotes
(3) Archaeobacteria (4) Both (2) and (3)

183. Which of the following is start codon ?

- (1) AUG (2) CUG
(3) UUG (4) GUG

184. Cistron does not has information related to

- (1) hn-RNA (2) m-RNA
(3) Protein (4) r-RNA

185. Elongation of nascent RNA requires :-

- (1) RNA polymerase + sigma factor
(2) RNA polymerase + rho factor
(3) Only RNA polymerase core enzyme
(4) RNA polymerase + sigma + rho factors

SECTION-B (BIOLOGY-II)

186. Which of the following does not synthesize r-RNA ?

- (1) RNA Poly-I
(2) RNA Poly-III
(3) Prokaryotic RNA polymerase
(4) RNA poly-II

187. **Assertion** : Louis Pasteur, in his experiment demonstrated that life comes only from pre-existing life.

Reason : In his experiment life did not come from killed yeast in pre-sterilized flask.

- (1) Both assertion & reason are true & the reason is a correct explanation of the assertion.
(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both assertion & reason are false.

188. Refer the given statements and select the correct ones.

- (i) Fossils are remains of hard parts of life forms in rocks.
(ii) A study of fossils in different sedimentary layers indicates the geological period in which they live.
(iii) Radio isotopes are often used to determine the age of the fossils.
(iv) Study of fossils is called palaeontology.

- (1) (i), (ii), (iii) and (iv)
(2) (ii) and (iv)
(3) (i), (iii) and (iv)
(4) None of these

189. Which of the following set in man includes vestigial organs?

- (1) Coccyx, vermiform appendix and ear muscles
(2) Body hair, atlas vertebra and ear muscles
(3) Coccyx, wisdom tooth and patella
(4) Body hair, cochlea, vermiform appendix and tongue

190. Match following evolution concepts in List-I with List-II and select the correct answer using the codes given below the lists:

List I		List II	
A.	Mutation	1.	changes in population's allele frequencies due to chance alone
B.	Gene flow	2.	differences in survival and reproduction among variant individuals
C.	Natural selection	3.	multiple migrations change allele frequencies.
D.	Genetic drift	4.	Source of new alleles

- (1) A-1, B-2, C-3, D-4
(2) A-4, B-2, C-3, D-1
(3) A-5, B-1, C-4, D-2
(4) A-4, B-3, C-2, D-1

191. Why is reproductive isolation important from an evolutionary standpoint ?
- (1) When the organisms of two populations of a species can no longer interbreed the flow of genetic material between them stops.
 - (2) It is not important from an evolutionary standpoint, the question is based on a false assumption.
 - (3) Reproductive isolation increases the mutation rate.
 - (4) Reproductive isolation may speed reproduction.
192. **Statement-I** : Evolution is a directed process in the sense of determinism.
Statement-II : Evolution is a stochastic process based on chance events in nature and chance mutation in organisms.
- (1) Statement I and II both are correct
 - (2) Statement I and II both are incorrect
 - (3) Only Statement I is correct
 - (4) Only Statement II is correct
193. Vertebrates 'hearts with respect to vertebrates' brains are :-
- (1) Homologous organs
 - (2) Analogous organs
 - (3) Vestigial organs
 - (4) Non-related organs
194. Which of the following was first to hunt with stone weapons ?
- (1) Homo habilis
 - (2) Homo erectus
 - (3) Australopithecus
 - (4) Neanderthal man
195. Industrial melanism includes which of the following ?
- (A) Natural selection
 - (B) Anthropogenic action
 - (C) Camouflage
 - (D) Directional selection
 - (E) Predation
 - (F) Pollution
- (1) A, B, C, D, E and F
 - (2) A, B, D, E and F
 - (3) A, B, E and F
 - (4) A, D and F
196. Which of the following serves dual purpose during DNA replication ?
- (1) Ribonucleotide triphosphate
 - (2) Ribonucleoside triphosphate
 - (3) Deoxyribonucleotide triphosphate
 - (4) Deoxyribonucleoside triphosphate
197. Which of the following are reasons behind stability of DNA over RNA ?
- (A) Lacks-OH groups on 2'
 - (B) Base pairing
 - (C) Lacks methylated uracil
 - (D) Stacking of base pairs
 - (E) Hydrogen bonds between base pairs
- (1) A, B, C, D and E (2) A, B, D and E
 - (3) A, B, C and D (4) A, B and D
198. **Assertion** : Eukaryotes have multiple ORI-sites
Reason : Eukaryotes have longer DNA than prokaryotes.
- (1) Both assertion and reason are true and reason is the correct explanation of assertion.
 - (2) Assertion is true but reason is false
 - (3) Both assertion and reason are false
 - (4) Both assertion and reason are true and reason is not the correct explanation of assertion

199. Triplet nature of genetic code was proved by :-

- (1) Nirenberg and Matthhai
- (2) George Gamow
- (3) Hargobind khorana
- (4) Severo-ocha

200. In eukaryotes, TATA box is present at :

- (1) 20 bp downstream
- (2) 20 bp upstream
- (3) 10 bp downstream
- (4) 10 bp upstream

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